

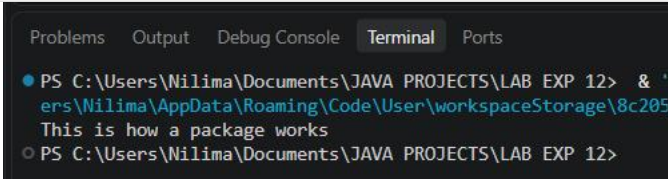


## Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

| Lab Assignment No.:                               |                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name of Student</b>                            | Akshit Gawade                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>PRN No.</b>                                    | 25070122029                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Batch</b>                                      | 2025-2029                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Class</b>                                      | A1                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Academic Year &amp; Semester</b>               | Second Year<br>Semester III                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Date of Submission</b>                         |                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Title of Assignment:</b>                       | Write a Java program to create and use user-defined packages                                                                                                                                                                                                                                                                                                                                                  |
| <b>Problem Statement &amp; Learning Objective</b> | <b>Problem Statement</b><br>Write a Java program to create and use user-defined packages.<br><b>Learning Objectives</b><br>To create and use user-defined packages for organizing Java programs                                                                                                                                                                                                               |
| <b>Theory</b>                                     | Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages. |
| <b>Program Code</b>                               | <pre>import package1.MyClass;  public class Main {     public static void main (String[] args) {         MyClass obj = new MyClass();         obj.display();     } }</pre><br><b>MyClass.java:</b>                                                                                                                                                                                                            |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <pre>package package1;  public class MyClass {     public void display () {         System.out.println("This is how a package works");     } }</pre>                                                                                                                                                                                                                                                                                                                                                                                                 |
| Output Screenshots    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Exercise Qns and Code | <ol style="list-style-type: none"> <li>1. Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.</li> <li>2. Create a Java program by developing a <i>user-defined package</i> named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.</li> </ol> |
| Output Screenshots    | <pre>EXERCISE 1  Ex1.java package Exercise.Ex1;  import Exercise.Ex1.student.Student; import Exercise.Ex1.faculty.Faculty;  public class Ex1 {     public static void main(String[] args) {         Student s = new Student(101, "Akshit", "Computer Science and Engineering");         Faculty f = new Faculty(201, "Dr. Dutta", "CSE");          s.display();         f.display();     } }  -----  Student.java package Exercise.Ex1.student;</pre>                                                                                                |

```
public class Student {  
    int id;  
    String name;  
    String course;
```

```
    public Student(int id, String name, String course) {  
        this.id = id;  
        this.name = name;  
        this.course = course;  
    }
```

```
    public void display() {  
        System.out.println("----- Student Details -----");  
        System.out.println("Student ID : " + id);  
        System.out.println("Name      : " + name);  
        System.out.println("Course       : " + course);  
    }  
}
```

Faculty.java

```
package Exercise.Ex1.faculty;
```

```
public class Faculty {  
    int id;  
    String name;  
    String department;
```

```
    public Faculty(int id, String name, String department) {  
        this.id = id;  
        this.name = name;  
        this.department = department;  
    }
```

```
    public void display() {  
        System.out.println("\n----- Faculty Details -----");  
        System.out.println("Faculty ID : " + id);  
        System.out.println("Name      : " + name);  
        System.out.println("Department : " + department);  
    }  
}
```

```

Problems Output Debug Console Terminal Ports
● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 12> & 'cd 'c:\Users\Nilima\AppData\Roaming\Code\User\workspace
s' '-cp' 'C:\Users\Nilima\AppData\Roaming\Code\User\workspace
----- Student Details -----
Student ID : 101
Name       : Akshit
Course     : Computer Science and Engineering

----- Faculty Details -----
Faculty ID : 201
Name       : Dr. Dutta
Department : CSE
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 12>

```

## EXERCISE 2:

```

● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 12> c::; cd 'c:\Users\Nilima\AppData\Roaming\Code\User\workspace\hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' 't.java\jdt_ws\LAB EXP 12_cc80a229\bin' 'Exercise.Ex2.Ex2'
----- Book Details -----
Book ID : 101
Title   : Wimpy Kid
Author  : Greg Heffley
Price   : Rs.400.0
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 12>

```

## CODE:

BOOK.java:

```

package Exercise.Ex2.library;

public class Book {
    int bookId;
    String title;
    String author;
    double price;

```

```

    public Book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
    }

```

```

    public void display() {
        System.out.println("----- Book Details -----");
        System.out.println("Book ID : " + bookId);
        System.out.println("Title   : " + title);
        System.out.println("Author  : " + author);
        System.out.println("Price   : Rs." + price);
    }
}

```

Ex2.java:

```

package Exercise.Ex2;

import Exercise.Ex2.library.Book;

```

|                   |                                                                                                                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <pre>public class Ex2 {<br/>    public static void main(String[] args) {<br/>        Book b = new Book(101, "Wimpy Kid", "Greg Heffley",<br/>400.00);<br/><br/>        b.display();<br/>    }<br/>}</pre>                                                                                                                      |
| <b>Conclusion</b> | <p>This experiment focuses on modular code organization and encapsulation through custom Java packages. By structuring classes into logical directory hierarchies, it illustrates how packages simplify namespace management, control accessibility, and streamline maintainability across large multi-component projects.</p> |